Problem 1

1) For Newton Raphson, we will need to minimize $l(\theta)$ and will need $l'(\theta)$ and $l''(\theta)$, where $\theta_{t+1} = \theta - \frac{l'(\theta)}{l''(\theta)}$ slide 9

I decided to run 9 iterations like the sample code, updating theta each time with the formula above.

$$l'(\theta) = -\sum (y - \exp(\theta x))(x \cdot \exp(\theta x))$$

$$l''(\theta) = \sum (x \exp(\theta x))^2 - (y - \exp(\theta x))(x^2 \exp(\theta x))$$

2) $\boxed{\hat{\theta} = 1.246822}$

```
> theta_hat
[1] 1.246822
```

```r
setwd("/Users/alwinlin/Desktop/DSC 382 Regression")
# Read data
data <- read.csv(file = 'HW7_data.csv', header = TRUE)
# Problem 1
y <- data$y
x <- data$x
n <- 100

# Initial value
t <- 1

# Newton-Raphson algorithm
for(k in 2:10){
  # First derivative of SSE
  u  <- -sum((y - exp(t * x)) * (x * exp(t * x)))
  # Second derivative
  d  <- sum((x * exp(t * x))^2 - (y - exp(t * x))*(x^2 * exp(t * x)))
  # Update theta
  t <- t - u/d
}
theta_hat <- t # This is for clarification that value 1.2468 is theta hat
```

3) For parametric bootstrap, we need to calculate parameters and for this question we will use y^*, $\theta^* = 1$ where $y^* = \exp(\hat{\theta} x) + \hat{\sigma}\, \text{rnorm}(100)$. Then we run the bootstrap 2:10 times like before to get our $\hat{\theta}_{boot}$ and the variance

4) $Var(\hat{\theta}_{boot}) = \boxed{4.745527 \times 10^{-5}}$

```
> var_theta
[1] 4.745527e-05
```

```r
y_hat <- exp(theta_hat * x)
sigma_hat <- sqrt(mean((y - y_hat)^2))
B <- 100
theta_boot <- numeric(B)
for(b in 1:B){
  y_star <- exp(theta_hat * x) + sigma_hat * rnorm(n) # Parametric bootstrap parameter
  t_star <- 1
  for(k in 2:10){
    u  <- -sum((y_star - exp(t_star * x)) * (x * exp(t_star * x)))
    d  <- sum((x * exp(t_star * x))^2 - (y_star - exp(t_star * x))*(x^2 * exp(t_star * x)))
    t_star <- t_star - u/d
  }
  theta_boot[b] <- t_star
}

# Bootstrap variance
var_theta <- var(theta_boot)
var_theta
```

5)

```
# Problem 5
x_bar <- mean(x)

log_yhat <- theta_hat * x_bar

# variance using delta method
var_logy <- x_bar^2 * var_theta
std_logy <- sqrt(var_logy)

CI_lower <- log_yhat - 1.96 * std_logy # 1.96 is z-score for 95% confidence
CI_upper <- log_yhat + 1.96 * std_logy

c(CI_lower, CI_upper)
```

```
> c(CI_lower, CI_upper)
[1] 0.7940882 0.8114751
```

95% between [0.7941, 0.8115]

Problem 2 Non linear $l(\theta) = \frac{1}{2}\sum_{i=1}^{n}(y_i - m(\theta, x_i))^2$

1) $l'(\theta) = \frac{1}{2}\sum_{i=1}^{n} 2(y_i - m_i(\theta))(-m_i'(\theta)) = -\sum_{i=1}^{n}(y_i - m_i(\theta))(m_i'(\theta))$

$y_i - m_i(\theta_0) = \sigma\varepsilon_i \rightarrow l'(\theta_0) = -\sigma\sum\varepsilon_i m_i'(\theta_0)$

$$\boxed{E[l'(\theta)] = -\sigma\sum m_i'(\theta_0)E[\varepsilon_i] = 0}$$

$l''(\theta) = \sum(m_i'(\theta))^2 - \sum(y_i - m_i(\theta))m_i''(\theta) = \sum(m_i'(\theta_0))^2 - \sum\sigma\varepsilon_i m_i''(\theta_0)$

$$\boxed{E[l''(\theta_0)] = \sum_{i=1}^{n}(m_i'(\theta_0))^2}$$

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2) $l'(\theta_0) = -\sigma\sum\varepsilon_i m_i'(\theta_0)$ $\quad Var(\sum\varepsilon_i m_i') = \sum(m_i'(\theta_0))^2$

$Var(l'(\theta_0)) = \sigma^2 Var(\sum\varepsilon_i m_i'(\theta_0))$

$Var(l'(\theta_0)) = \sigma^2 \underbrace{\sum(m_i'(\theta_0))^2}_{E[l''(\theta_0)]}$

$$\boxed{Var(l'(\theta_0)) = \sigma^2 E[l''(\theta_0)]}$$

3) $l'(\theta_0) \approx \cancel{l'(\hat{\theta})}^{\,0} + (\theta_0 - \hat{\theta})\, l''(\hat{\theta})$ $\hat{\theta}$ is minimized so that $l'(\hat{\theta}) = 0$

$l'(\theta_0) \approx (\theta_0 - \hat{\theta})\, C$, where $C = l''(\hat{\theta})$, solve for $\hat{\theta}$

$$\boxed{\hat{\theta} \approx \theta_0 - \frac{l'(\theta_0)}{C}}$$

---

4) For mean, $\hat{\theta} \approx \theta_0 - \cancel{\frac{l'(\theta_0)}{C}}^{\,0}$, where $E[l'(\theta)] = 0$. So mean is θ_0

$$Var(\hat{\theta}) = \frac{Var(l'(\theta_0))}{C^2} = \frac{\sigma^2\, \cancel{E[l''(\theta_0)]}}{C^{\cancel{2}}}, \text{ where } C = E[l''(\theta_0)]$$

$Var(\hat{\theta}) = \dfrac{\sigma^2}{E[l''(\theta_0)]}$ and $E[l''(\theta_0)] = \sum (m_i'(\theta_0))^2$

$$\boxed{\hat{\theta} \approx N\left(\theta_0, \frac{\sigma^2}{\sum_{i=1}^{n} (m_i'(\theta_0))^2}\right)}$$

---

5) $$Std(\hat{\theta}) = \sqrt{\frac{\sigma^2}{\sum_{i=1}^{n} (m_i'(\theta_0))^2}} = \frac{\sigma}{\sqrt{\sum_{i=1}^{n} (m_i'(\theta_0))^2}}$$

$$\boxed{\left(\hat{\theta} - Z_{\alpha/2} \frac{\sigma}{\sqrt{\sum_{i=1}^{n} (m_i'(\theta_0))^2}},\ \hat{\theta} + Z_{\alpha/2} \frac{\sigma}{\sqrt{\sum_{i=1}^{n} (m_i'(\theta_0))^2}}\right)}$$